

Exam-Ready Notes: Introduction to Philosophy, Logic & Critical Thinking

MODULE 1: WHAT IS PHILOSOPHY?

The Meaning of Philosophy

The word "philosophy" comes from two Greek words: *philo* (love) and *sophia* (wisdom). So philosophy is literally the love of wisdom. It is the oldest surviving intellectual discipline in the Western tradition, and almost every other subject we study today — biology, physics, psychology, economics, even AI — branched off from philosophy at some point in history, as certain questions got answered satisfactorily and became their own field. What philosophy does is *stick with* the unanswered questions.

A common misconception is that philosophy is an esoteric, impractical subject for specialists. In fact, as Stich and Donaldson point out, philosophical questions come up in everyday life all the time. Consider whether it is okay to eat meat. That question quickly leads you into questions about animal consciousness, the relationship between pain and the nervous system, the existence of God, and the reliability of scripture — all of which are philosophical questions. Philosophy is everywhere, even if we don't always recognize it.

What Makes a Question Philosophical?

There are two useful ways to understand what makes a question philosophical.

First, philosophy is especially concerned with **normative** questions — questions about what *ought to be*, not just what *is*. A normative question asks about right and wrong, good and bad, how we should live, how society should be structured. "How many people live in Mumbai?" is a descriptive question. "Should voting be compulsory?" is a normative one — and a philosophical one. Science can tell us facts about the world (descriptive), but philosophy asks how we should evaluate those facts and what we should do with them.

Second, philosophy is also in the business of **questioning assumptions** and **clarifying concepts**. Philosophers scrutinize the basic assumptions of science, politics, art, religion — everything. This is why Socrates was put on trial in ancient Athens and condemned to death for "corrupting the youth." He made people question what they thought they already knew, which was seen as dangerous. This is also why philosophy has sometimes been regarded as a subversive activity.

Some example philosophical questions: Is it possible to travel backward in time? Do we have free will? What is the difference between knowledge and opinion? Could a digital computer be conscious? When, if ever, is it okay to lie? These are not esoteric puzzles — everyone confronts them sooner or later.

The Philosophical Method

Philosophers do not just give opinions. What separates philosophy from mere opinion-sharing is

that philosophers attempt to state their views **clearly and precisely**, and they give **explicit arguments** for those views. In 1948, the philosophers Frederick Coppleston and Bertrand Russell debated the existence of God on radio. What is striking about their exchange is not that they disagreed — it's how they disagreed. They first agreed on a precise definition of "God." Coppleston then stated exactly what he was claiming: that God exists and that this can be proved philosophically. Russell stated exactly what he was not claiming: he was not an atheist (someone who says God definitely does not exist), but an agnostic (someone who says it cannot be proved either way). This kind of precise articulation of disagreement is the philosophical method.

Importantly, philosophers do not defend their views by appealing to authority alone. "Aristotle said so" or "The scripture says so" is not an argument. Philosophy expects you to provide reasons — premises that give someone genuine justification to believe your conclusion.

Philosophy and Science

Some scientists, like Stephen Hawking, have claimed that philosophy is now dead because science has taken over. Hawking said: "Traditionally these are questions for philosophy, but philosophy is dead... Scientists have become the bearers of the torch of discovery." Stich and Donaldson disagree, and rightly so: there are many philosophical questions that science, as currently practised, does not answer. No amount of neuroscience will tell you whether you have free will. No zoology journal will tell you whether it is morally okay to eat fish. These questions are currently outside the scope of science, even if they might one day be addressed scientifically.

Why Study Philosophy?

Bertrand Russell gave a beautiful answer: Philosophy "removes the somewhat arrogant dogmatism of those who have never travelled into the region of liberating doubt, and it keeps alive our sense of wonder by showing familiar things in an unfamiliar aspect." Hilary Putnam's more blunt answer was: "Because bad philosophy is all around!" Philosophy trains you to think clearly, to realise what you don't know, to understand other people's views, and to reason well. It is also an antidote to the **Dunning-Kruger effect** — the tendency of people to overestimate their knowledge in areas where they are actually ignorant. The more ignorant we are, the more confident we tend to be. Philosophy introduces what Socrates called **Socratic irony** — "All I know is that I know nothing" — as a corrective to that overconfidence.

Key Domains of Philosophy

- **Logic:** What is right reasoning?
 - **Epistemology:** How is knowledge possible?
 - **Ontology/Metaphysics:** What exists? What is the relation between mind and world?
 - **Ethics:** What is right or wrong? Good or bad?
 - **Axiology:** What is value?
 - **Aesthetics:** What is beauty?
 - **Philosophy of X:** Philosophy of law, science, economics, sport, AI, etc.
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MODULE 2: LOGIC AND CRITICAL THINKING

What is Logic?

Logic is the study of the methods and principles used to distinguish **correct** from **incorrect** reasoning. It is a normative discipline: it tells us not how people actually reason (that is psychology, a descriptive science), but how they *ought* to reason. Logic is, in a sense, the art of self-defence — against bad arguments, rhetoric, propaganda, and manipulation.

We make inferences all the time. When we hear about AI job losses and conclude AI is responsible for tech layoffs, that is an inference. When we see a product has many five-star reviews and infer it is good, that is an inference. The question is whether these inferences are good ones. Logic gives us the tools to answer that.

Propositions and Arguments

The building blocks of logic are **propositions** (also called statements). A proposition is any claim that asserts something is or is not the case, and therefore has a **truth value** — it is either true or false. "It is raining," "All dogs are mammals," and "Mumbai is the capital of India" are all propositions (though the last one is false). Questions, commands, and exclamations are not propositions — they don't assert anything that can be true or false.

An important distinction: a proposition is not the same as a sentence. The same proposition can be expressed by different sentences in different languages or wordings. "It is raining," "Il pleut" (French), and "Es regnet" (German) all express the same proposition.

An **argument** is a cluster of propositions in which some (the **premises**) are offered as reasons to believe another (the **conclusion**). Think of it like a building: the floor is the premises, the roof is the conclusion, and the cement holding them together is the **inference**. The foundation — what makes the whole structure stable — is the laws of logic themselves.

Laws of Thought

There are three foundational laws of logic:

1. **The Law of Non-Contradiction:** Nothing can both have a property and lack it at the same time. A thing cannot be both A and not-A simultaneously.
2. **The Law of Identity:** Everything is identical to itself. A is A.
3. **The Law of Excluded Middle:** For any property, everything either has it or lacks it. Either A or not-A — there is no third option.

These are called the laws of *thought* because without them, thought and communication would be impossible. Denying the law of non-contradiction, for instance, is self-defeating — the very act of denying it affirms it. This is an instance of **reductio ad absurdum**: a form of argument in which you show that denying something leads to an absurd or impossible conclusion.

Truth vs. Validity

This is one of the most important distinctions in logic, and it is frequently confused.

Truth is a property of **propositions**. A proposition is true if it matches the way the world actually is. "The sun rises in the east" is true. "Mumbai is the capital of India" is false.

Validity is a property of **arguments**. An argument is valid if the conclusion *necessarily follows* from the premises — that is, if it is impossible for all the premises to be true and the conclusion to be false at the same time. Arguments cannot be called true or false. Propositions cannot be called valid or invalid. This is a category error people make all the time.

Consider: "All women are mortal. Socrates is a woman. Therefore, Socrates is mortal." This argument is **valid** — the conclusion does follow from the premises. But the second premise is false. So the argument is valid but not sound.

Soundness requires *both* validity *and* all true premises. A sound argument is one where the reasoning is valid AND all the premises are actually true. Only a sound argument can definitively establish the truth of its conclusion.

Here is a map of the possible combinations (from Copi and Cohen):

	True Conclusion	False Conclusion
Valid Arguments	Possible	Possible (if a premise is false)
Invalid Arguments	Possible	Possible

The critical point is: the only combination that **cannot** exist is a valid argument with all true premises and a false conclusion. If an argument is valid and the premises are all true, the conclusion *must* be true.

Logical vs. Physical Impossibility

To understand deductive validity deeply, we need this distinction.

Physical impossibility means something is inconsistent with the laws of nature. Science tells us what is physically possible. It is physically impossible for a human being to run at the speed of sound, for instance.

Logical impossibility means something is inconsistent with the laws of logic itself. A "married bachelor" is logically impossible — it is self-contradictory by definition. A "square circle" is logically impossible. These are not just rare or difficult — they cannot exist in any conceivable world. All oxymorons ("less is more," "same difference") are really logical near-impossibilities.

The relationship between the two: **whatever is physically possible is logically possible**, but not everything that is logically possible is physically possible. Archimedes said, "Give me a lever long enough and a fulcrum on which to place it, and I shall move the world." That is logically

possible (there is no contradiction in the idea), but physically impossible (no such lever and fulcrum exist). There is nothing that is physically possible but logically impossible.

A deductive argument is **truth-preserving** if it is not logically possible for all its premises to be true and its conclusion false. This is what validity means in deductive logic.

Types of Inferences: Deductive vs. Inductive

Deductive arguments claim that if the premises are true, the conclusion *must* be true — with absolute certainty. They are truth-preserving. The conclusion is, so to speak, already hidden within the premises. Because of this, deductive arguments are also called **non-ampliative** — they don't add new information to what the premises already contain. If a valid deductive argument has a false conclusion, you know with certainty that at least one premise must be false.

Inductive arguments claim only that the premises make the conclusion *probable* — not certain. They are **ampliative**: they go beyond what the premises strictly contain. Every single swan I have ever seen is white, therefore all swans are white. This might seem reasonable, but it was refuted when black swans were discovered in Australia. Inductive arguments can always be made stronger or weaker by additional information — they are **non-monotonic**.

Inductive arguments are evaluated on a spectrum of **strength** (strong to weak), not by the binary of valid/invalid. Empirical science relies heavily on inductive inference: from observed samples to general conclusions, from correlations to causal generalizations.

One deep problem with induction is its **circularity**. What makes inductive inference legitimate? The principle of the uniformity of nature — the idea that the future will be like the past. But what justifies *that* principle? We would need to say: "In the past, the future has always been like the past — therefore it will continue to be so in the future." But this is just using induction to justify induction. As David Hume first pointed out, this is circular reasoning.

Modus Ponens and Modus Tollens

These are the two fundamental valid forms of conditional (if-then) reasoning.

Modus Ponens (valid form):

- If A, then B
- A (the antecedent is *affirmed*)
- Therefore, B

The **fallacious** form of this is **affirming the consequent**:

- If A, then B
- B
- Therefore, A (INVALID)

Modus Tollens (valid form):

- If A, then B

- Not-B (the consequent is *denied*)
- Therefore, Not-A

The **fallacious** form of this is **denying the antecedent**:

- If A, then B
- Not-A
- Therefore, Not-B (INVALID)

Necessary and Sufficient Conditions

In a conditional statement "If A, then B":

- **A is a sufficient condition for B:** A's presence guarantees B. If there's fire, there's oxygen. Fire is sufficient for oxygen.
- **B is a necessary condition for A:** B must be present for A to occur. Oxygen is necessary for fire. Without oxygen, no fire.

Examples to remember: Oxygen is necessary for combustion. Combustion is sufficient to prove the presence of oxygen. Having fuel is necessary for an engine to run. Holding a guitar is necessary to be a guitarist, but not sufficient (you also have to know how to play).

MODULE 3: INFORMAL FALLACIES

What is a Fallacy?

A fallacy is a type of argument that *seems* correct but, on examination, is not. More precisely, a fallacy is an identifiable pattern of incorrect reasoning — the premises appear to support the conclusion but, in fact, do not. The key word is "pattern" — a fallacy is not just any one-off mistake, but a recognizable type of mistake that appears again and again.

A distinction worth noting: **sophistry** is the deliberate use of illogical or misleading arguments by someone who knows they are wrong. A **fallacy**, by contrast, may be committed innocently, without the speaker realising the error. The term "fallacy" carries no implication of bad intent.

An important caution: pointing out that someone has made a *bad argument* for a conclusion does not mean the conclusion is false. A true conclusion can be supported by a terrible argument. If a bad argument fails, it only means those specific premises don't justify that conclusion — there might be another set of premises that do.

Classification of Fallacies

Informal fallacies are grouped into four main types:

1. **Fallacies of Relevance** — premises are simply not relevant to the conclusion

2. **Fallacies of Defective Induction** — premises are relevant but too weak or insufficient
 3. **Fallacies of Presumption** — premises assume something that hasn't been proved
 4. **Fallacies of Ambiguity** — mistakes arise from unclear or shifting word meanings
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FALLACIES OF RELEVANCE

These are the most common fallacies. The premises have no real logical connection to the conclusion, even though they may seem emotionally or psychologically persuasive.

R1. Appeal to the Populace (Ad Populum) This fallacy occurs when, instead of giving rational evidence, someone wins assent by stirring up popular feelings. The argument is: "everyone believes X / everyone is doing X, therefore X is right." The classic variant is the **bandwagon fallacy** — buy this product because everyone is buying it. The fact that many people believe something gives us no logical reason to think it is true. Propaganda and advertising rely heavily on this fallacy.

R2. Appeal to Emotion (Ad Misericordiam — Appeal to Pity) Here, the arguer appeals to emotions like pity, fear, or guilt rather than providing rational reasons. Appeal to pity (ad misericordiam) is particularly common in legal contexts — a defense attorney might describe a client's terrible childhood to win acquittal, even when the question at issue is simply whether the client committed the crime. Pity is not irrelevant to sentencing, but it is irrelevant to the question of guilt.

R3. Red Herring A red herring is when the arguer deliberately changes the subject to a different (but perhaps related) issue, draws a conclusion about the new issue, and implies this settles the original dispute. Also called "whataboutism." Example: "Environmentalists worry about nuclear power. But electricity is dangerous in general — hundreds are electrocuted every year due to carelessness. If people just exercised caution..." The original topic (nuclear power's specific dangers) has been abandoned for a different topic (electrical safety in general).

R4. Straw Man The straw man fallacy occurs when an arguer misrepresents an opponent's position — making it more extreme or easier to attack than it actually is — and then attacks this distorted version. Watch for phrases like "so what you're saying is..." or "the real reason you believe this is..." A related idea is the **principle of charity**: in good philosophical discourse, you should represent your opponent's position in its strongest form, not its weakest. Straw man arguments sometimes backfire — if the distortion is obvious, audiences may sympathise with the misrepresented party.

R5. Attack on the Person (Ad Hominem) Rather than addressing the argument, the attacker goes after the person making it. There are two main forms:

Abusive ad hominem: Attacking the character, intelligence, or integrity of the opponent. "Bill Maher argues that religion is foolish, but he's an arrogant pig, so his arguments aren't worth listening to." The character of a person is logically irrelevant to whether their argument is sound. Variants include **character assassination**, **guilt by association**, and **poisoning the well**

(attacking someone's credibility in advance so that everything they say will be dismissed before being heard).

Circumstantial ad hominem: Arguing that a person must accept or reject some view because of their personal circumstances (their profession, nationality, religion, etc.). A special subtype is **tu quoque** ("you did it too") — responding to a charge by pointing out that the accuser does the same thing. Whether or not the accuser is guilty is irrelevant to the truth of the original charge.

R6. Appeal to Force (Ad Baculum) This fallacy involves using threats, coercion, or force to win acceptance of a conclusion. "Ad baculum" means "appeal to the stick." The threatened party may comply, but compliance achieved by force is not rational persuasion. The appeal to force is the abandonment of reason.

R7. Missing the Point (Ignoratio Elenchi) Also called "irrelevant conclusion," this fallacy occurs when the premises genuinely support *some* conclusion — just not the conclusion being drawn. The arguer reaches a different, vaguely related conclusion. Example: "Crimes have been increasing. Therefore we must reinstate the death penalty immediately." The evidence supports some action on crime, but does not specifically support the conclusion about the death penalty. Every fallacy of irrelevance is in a way a form of ignoratio elenchi, but we reserve the name for cases that don't neatly fit other categories. The related term **non sequitur** ("does not follow") is used informally when the gap between premises and conclusion is glaringly obvious.

FALLACIES OF DEFECTIVE INDUCTION

In these fallacies, the premises are not completely irrelevant, but they are far too weak to support the conclusion.

D1. Argument from Ignorance (Ad Ignorantiam) Arguing that something is true because it has not been proved false, or false because it has not been proved true. Just because we can't prove UFOs don't exist doesn't mean they do. Just because we can't prove a new drug is harmful doesn't mean it is safe. Absence of evidence is not always evidence of absence. In some careful scientific contexts (like drug testing), the absence of harmful effects after rigorous testing can count as some evidence of safety — but this is different from just claiming no one has disproved something.

D2. Appeal to Inappropriate Authority (Ad Verecundiam) We legitimately appeal to experts all the time, and this is not fallacious. The fallacy arises when the authority appealed to has no genuine expertise in the relevant field, or when one authority is appealed to in a domain where experts disagree. A famous actor endorsing a medication, or a physicist giving opinions on economic policy, are examples. We must ask: is this person genuinely expert in the matter at hand?

D3. False Cause (Non Causa Pro Causa) Incorrectly treating something as the cause of an event when it is not really the cause. The most common version is **post hoc ergo propter hoc** — "after this, therefore because of this." Just because B followed A does not mean A caused B. The sun rises every morning after people sleep — sleep does not cause the sunrise. Correlation is not causation.

A specific variant is the **slippery slope fallacy**: arguing that taking one step in a direction will necessarily lead to a chain of increasingly bad outcomes. "If we allow assisted suicide, doctors will soon be killing patients without consent." The problem is that the feared cascade of consequences is simply asserted, not demonstrated. Each step on the slope would require its own separate argument.

D4. Hasty Generalization (Converse Accident) Drawing a broad general conclusion from too few cases. Stereotypes are a common example — judging all people from a country by one or two experiences. To generalise responsibly, you need a sufficiently large and representative sample. Moving from one case (or a very few cases) to a sweeping universal claim is defective inductive reasoning.

FALLACIES OF PRESUMPTION

These fallacies smuggle in an unwarranted or unjustified assumption, which the argument depends upon but never proves.

P1. Accident This is the reverse of hasty generalization. Instead of moving incorrectly from a few cases to a general rule, accident involves incorrectly applying a general rule to a particular case where the rule doesn't apply because of special ("accidental") circumstances. Almost every rule has exceptions — ignoring those exceptions commits the fallacy of accident. "You should always keep your promises" is a good general rule, but it does not apply if keeping a promise requires you to harm someone.

P2. Complex Question (Plurium Interrogationum) Asking a question in a way that presupposes the truth of some conclusion buried within it. The classic example is "Have you stopped beating your wife?" — both "yes" and "no" imply you were beating her at some point. A question like "With all the hysteria and phony science, isn't man-made global warming the greatest hoax ever?" presupposes that the science is phony and that there is hysteria — assumptions that need to be argued for, not buried in a question. The trap is that answering the question seems to confirm the assumptions.

P3. Begging the Question (Petitio Principii) Also called circular argument, this fallacy assumes the truth of the very thing one is trying to prove. The conclusion is hidden in one of the premises, often disguised by different words. "Freedom of speech must be advantageous to the state, because it is conducive to the community's interests for individuals to enjoy unlimited freedom of expression." This says: freedom of speech is good because freedom of speech is good. The argument goes around in a circle. Importantly, circular arguments are always technically valid (the conclusion does follow from the premises), but they are worthless — they prove nothing that wasn't already assumed.

FALLACIES OF AMBIGUITY

These fallacies arise because words or phrases shift meaning within the argument.

A1. Equivocation Using the same word with two different meanings in the same argument. "The sign said 'fine for parking here,' so I parked." The word "fine" means both a penalty and "acceptable." Or: "All laws have a lawmaker; nature has laws; therefore nature has a lawmaker." The word "law" is being used in two different senses (legal laws vs. scientific laws). Equivocation is especially tricky with **relative terms** — words like "tall," "good," or "small" that mean different things in different contexts. A "good general" and a "good president" are different kinds of good.

A2. Amphiboly The fallacy arises when the grammatical construction of a sentence makes it ambiguous — it can be interpreted in two different ways. "Women prefer Democrats to men" — do they prefer Democrats over men, or do they prefer Democrats as opposed to preferring Republicans? The argument takes one interpretation as a premise and draws a conclusion that only follows from the other interpretation.

A3. Accent Meaning is distorted by a misleading emphasis on certain words or by taking words out of context. A theater critic who writes "it is far from the funniest play on Broadway" might find her words quoted as "funniest play on Broadway!" in an advertisement. Accent is especially common in political propaganda, advertising (large print offers with "and up" in tiny print), and out-of-context quotation.

A4. Composition Reasoning incorrectly from what is true of the *parts* to what is true of the *whole*. Each part of a machine is light in weight; therefore the whole machine is light. Or: each individual musician in an orchestra is skilled; therefore the orchestra as a whole will perform brilliantly. These conclusions don't necessarily follow — the whole can have properties different from its parts.

A5. Division The reverse of composition — reasoning incorrectly from what is true of the *whole* to what is true of the *parts*. The orchestra is world-famous; therefore every individual musician in it is world-famous. The university is excellent; therefore every department in it is excellent. What is true collectively of a group is not necessarily true of each of its members individually.

How to Identify a Fallacy

When reading or listening to an argument, follow these steps:

1. Identify the **conclusion** — what is the argument trying to establish?
2. Identify the **premises** — what reasons are being offered?
3. Ask: do the premises independently and genuinely support the conclusion?
4. If not, it is a fallacy. To identify which one, ask:
 - Are the premises just restating the conclusion in other words? → Begging the question
 - Are the premises attacking the person rather than the argument? → Ad hominem
 - Are the premises appealing to emotion, popularity, or force? → Fallacies of relevance
 - Are the premises based on too few cases? → Hasty generalization
 - Is a word being used in two different senses? → Equivocation

- Is a false or misleading causal connection being assumed? → False cause
 - Is the conclusion just different from what the premises actually establish? → Missing the point
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SUMMARY OF KEY DEFINITIONS

Term	Definition
Proposition	A statement that is either true or false
Argument	Premises offered in support of a conclusion
Valid argument	An argument where the conclusion necessarily follows from the premises
Sound argument	A valid argument with all true premises
Deductive argument	Conclusion follows with certainty if premises are true
Inductive argument	Conclusion follows with probability; can be stronger or weaker
Logical impossibility	Something inconsistent with the laws of logic (e.g. a square circle)
Physical impossibility	Something inconsistent with the laws of nature
Fallacy	A pattern of incorrect reasoning that seems correct
Modus Ponens	If A then B; A; therefore B (valid)
Modus Tollens	If A then B; not-B; therefore not-A (valid)
Affirming the Consequent	If A then B; B; therefore A (INVALID)
Denying the Antecedent	If A then B; not-A; therefore not-B (INVALID)
Sufficient condition	A guarantees B; A is sufficient for B
Necessary condition	B must be present for A; B is necessary for A

These notes cover all major topics from the lecture slides and assigned readings. For deeper understanding, refer to Copi and Cohen's Introduction to Logic and Stich and Donaldson's Philosophy: Asking Questions, Seeking Answers.